MSG No. 217.80 $I\bar{4}'3m'$ [Type III, cubic]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'3m'`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `6b`, site symmetry: `-4'2.m'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,13,14,21,22]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,8,10,12,17,18,19,20]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,9,11,15,16,23,24]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[25,26,27,28,37,38,45,46]`
5	$\left[0, \frac{1}{2}, 0\right]$	`[29,32,34,36,41,42,43,44]`
6	$\left[0, 0, \frac{1}{2}\right]$	`[30,31,33,35,39,40,47,48]`

Table 3: Wyckoff site: `8c`, site symmetry: `.3m'`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6,20,22,24]`
2	$[x, -x, -x]$	`[2,10,11,16,17,21]`
3	$[-x, x, -x]$	`[3,7,12,13,18,23]`
4	$[-x, -x, x]$	`[4,8,9,14,15,19]`
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[25,29,30,44,46,48]`
6	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[26,34,35,40,41,45]`
7	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	`[27,31,36,37,42,47]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	`[28,32,33,38,39,43]`

Table 4: Wyckoff site: `12d`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,2,37,38]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[3,4,45,46]`
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,12,41,43]`
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[6,9,40,47]`
5	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[7,11,39,48]`
6	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[8,10,42,44]`
7	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[13,14,25,26]`
8	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[15,24,31,35]`
9	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[16,23,30,33]`
10	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[17,19,29,36]`
11	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[18,20,32,34]`
12	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[21,22,27,28]`

Table 5: Wyckoff site: `12e`, site symmetry: `2.m'm'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2,21,22]`
2	$[-x,\ 0,\ 0]$	`[3,4,13,14]`
3	$[0,\ x,\ 0]$	`[5,12,18,20]`
4	$[0,\ 0,\ x]$	`[6,9,15,24]`
5	$[0,\ 0,\ -x]$	`[7,11,16,23]`
6	$[0,\ -x,\ 0]$	`[8,10,17,19]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[25,26,45,46]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[27,28,37,38]`
9	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[29,36,42,44]`
10	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[30,33,39,48]`
11	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[31,35,40,47]`
12	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[32,34,41,43]`

Table 6: Wyckoff site: `24f`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,2]`
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,4]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,12]`
4	$\left[\frac{1}{2},\ 0,\ x\right]$	`[6,9]`
5	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[7,11]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[8,10]`
7	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[13,14]`
8	$\left[0,\ \frac{1}{2},\ x\right]$	`[15,24]`
9	$\left[0,\ \frac{1}{2},\ -x\right]$	`[16,23]`
10	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[17,19]`
11	$\left[\frac{1}{2},\ x,\ 0\right]$	`[18,20]`
12	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[21,22]`
13	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[25,26]`
14	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[27,28]`
15	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[29,36]`
16	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[30,33]`
17	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[31,35]`
18	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[32,34]`
19	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[37,38]`
20	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	`[39,48]`
21	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	`[40,47]`
22	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[41,43]`
23	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[42,44]`
24	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[45,46]`

Table 7: Wyckoff site: `24g`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,20]`
2	$[x,\ -x,\ -z]$	`[2,17]`
3	$[-x,\ x,\ -z]$	`[3,18]`
4	$[-x,\ -x,\ z]$	`[4,19]`
5	$[z,\ x,\ x]$	`[5,24]`
6	$[x,\ z,\ x]$	`[6,22]`
7	$[-x,\ z,\ -x]$	`[7,13]`
8	$[-z,\ -x,\ x]$	`[8,15]`
9	$[-x,\ -z,\ x]$	`[9,14]`
10	$[z,\ -x,\ -x]$	`[10,16]`
11	$[x,\ -z,\ -x]$	`[11,21]`
12	$[-z,\ x,\ -x]$	`[12,23]`
13	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[25,44]`
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[26,41]`
15	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[27,42]`
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[28,43]`
17	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[29,48]`
18	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[30,46]`
19	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[31,37]`
20	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[32,39]`
21	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	`[33,38]`
22	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[34,40]`
23	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	`[35,45]`
24	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[36,47]`

Table 8: Wyckoff site: `48h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[y,\ z,\ x]$	`[6]`
7	$[-y,\ z,\ -x]$	`[7]`
8	$[-z,\ -x,\ y]$	`[8]`
9	$[-y,\ -z,\ x]$	`[9]`
10	$[z,\ -x,\ -y]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-z,\ x,\ -y]$	`[12]`
13	$[-x,\ z,\ -y]$	`[13]`
14	$[-x,\ -z,\ y]$	`[14]`
15	$[-z,\ -y,\ x]$	`[15]`
16	$[z,\ -y,\ -x]$	`[16]`
17	$[y,\ -x,\ -z]$	`[17]`

continued ...

Table 8

No.	position	mapping
18	$[-y,\, x,\, -z]$	[18]
19	$[-y,\, -x,\, z]$	[19]
20	$[y,\, x,\, z]$	[20]
21	$[x,\, -z,\, -y]$	[21]
22	$[x,\, z,\, y]$	[22]
23	$[-z,\, y,\, -x]$	[23]
24	$[z,\, y,\, x]$	[24]
25	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[29]
30	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[30]
31	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[31]
32	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[32]
33	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[33]
34	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[34]
35	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[35]
36	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[36]
37	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	[37]
38	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	[38]
39	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	[39]
40	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	[40]
41	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[41]
42	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[42]
43	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[43]
44	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[44]
45	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[45]
46	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[46]
47	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	[47]
48	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	[48]